Synthetic Native Template

Redistributable test fixture — not laboratory data

Representative content layout

- Native title/content placeholders

Progress / Previous Commitments

Commitment | NS101 | planned | Owner: Gary | Due: 2026-09-10T09:00:00Z
Commitment | NS201 | planned | Owner: Gary | Due: 2026-09-24T00:00:00Z

H002

Dependencies | pressure fixture | Parallel: True
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H001 │ Hypothesis

Hypothesis │ Bulk conductivity drives the positional signal gradient.
Falsifier │ Matched conditions do not change the predicted CV or resistance.
Research question │提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 │ Problem

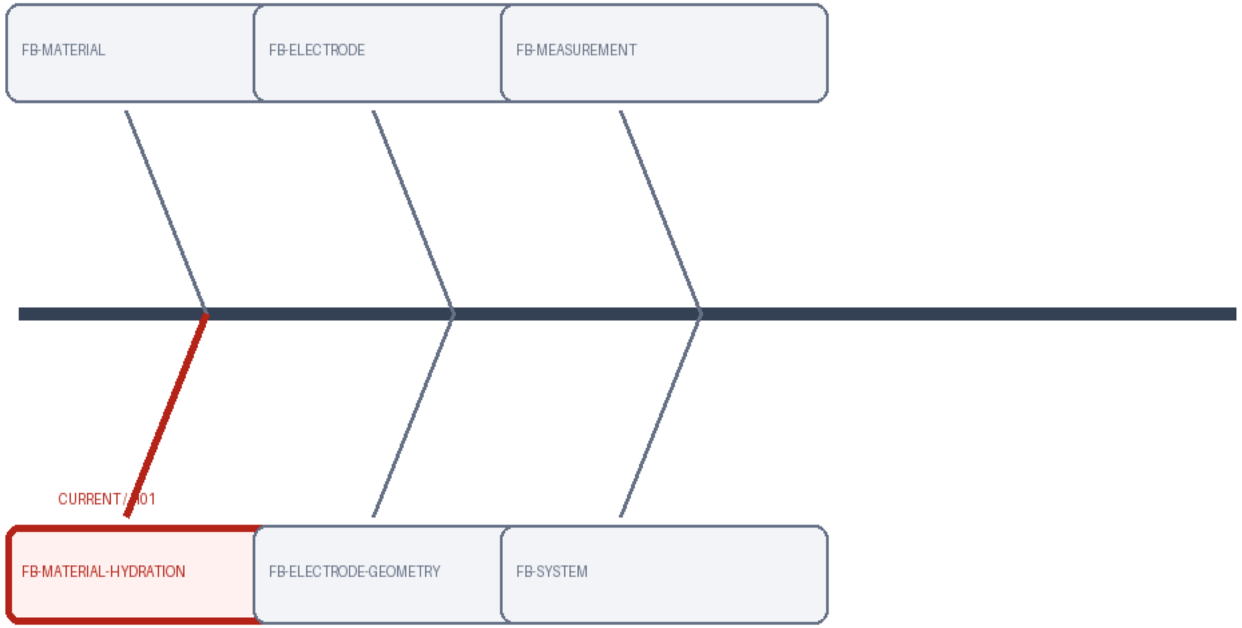
含水量與導電度存在位置梯度

導電度提高是否足以修復重複性仍未知。

提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 | Total Fishbone / Research Map

FB-MATERIAL-HYDRATION



H001 │ Observation → Literature → Mechanism

提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

Observation │位置依賴缺陷與訊號變異已觀察到。
Problem │現有模型無法解釋此變異。

H001 ｜ Literature → Mechanism → Strategy

Literature consensus ｜ transport gradient 可產生位置效應。
Alternatives ｜ interface contact remains alternative
Gap ｜缺少控制比較隔離機制。
Implication ｜需匹配條件後比較。

Mechanism ｜ bulk transport
Strategy ｜均化導電度

H001 │ Experiment Design

EXP101 │ IV: [' 含水量 ']
Controls: [' 原始配方 ']
N/replicates: {'replicates': 3, 'samples': 15}
Metrics/units: [' 導電度 (mS/cm)']
Method: ['synthetic-four-probe']

Prediction: [' 導電度提升 ']
Decision rule: {'go': 'mean conductivity increases >=20%', 'partial_go': '10-20%', 'no_go': '<10%'}

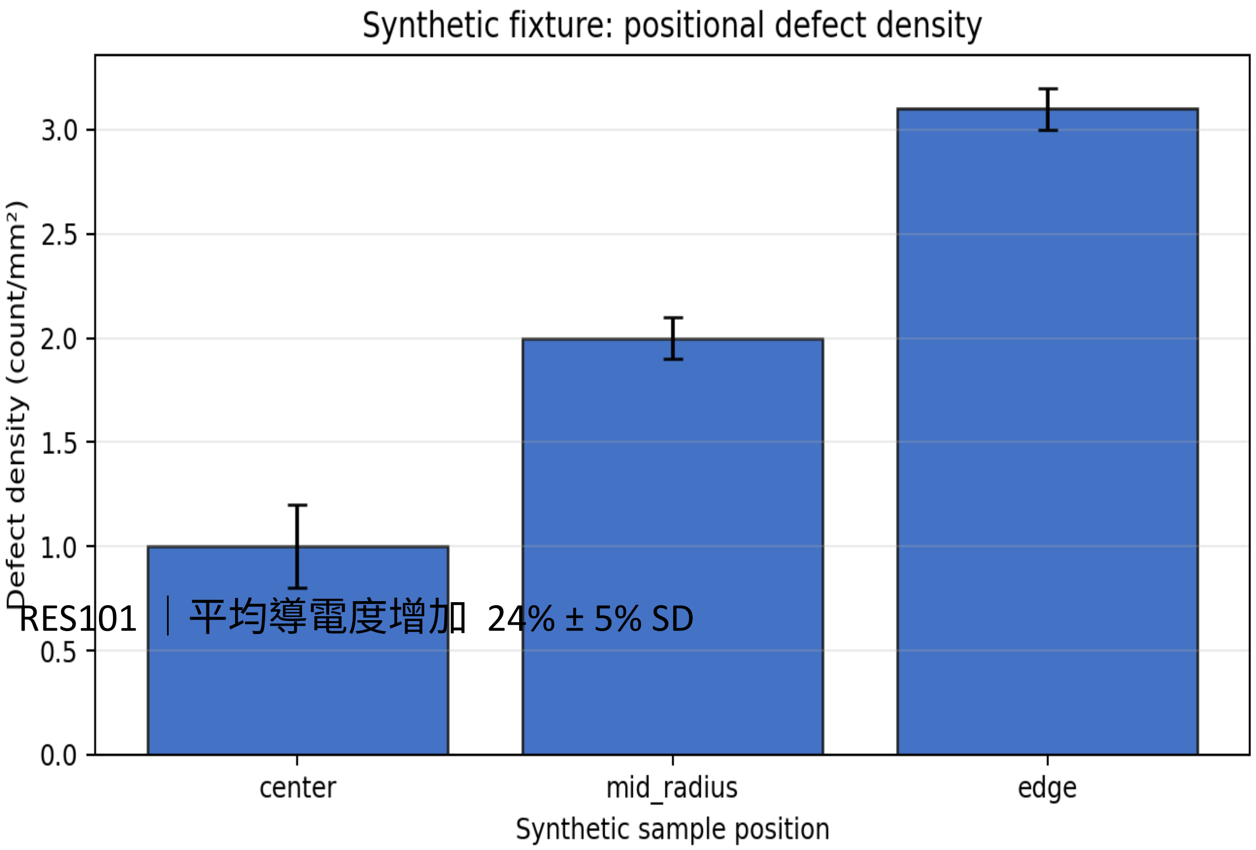
H001 │ Experiment Design

EXP102 │ IV: [' 位置 ']
Controls: [' 原始位置分布 ']
N/replicates: {'replicates': 3, 'samples': 15}
Metrics/units: [' 訊號 CV (%)']
Method: ['synthetic-signal-test']

Prediction: ['CV 下降 ']
Decision rule: {'go': 'CV decreases >=30%', 'partial_go': '10-30%', 'no_go': '<10%'}

H001 ｜ Result

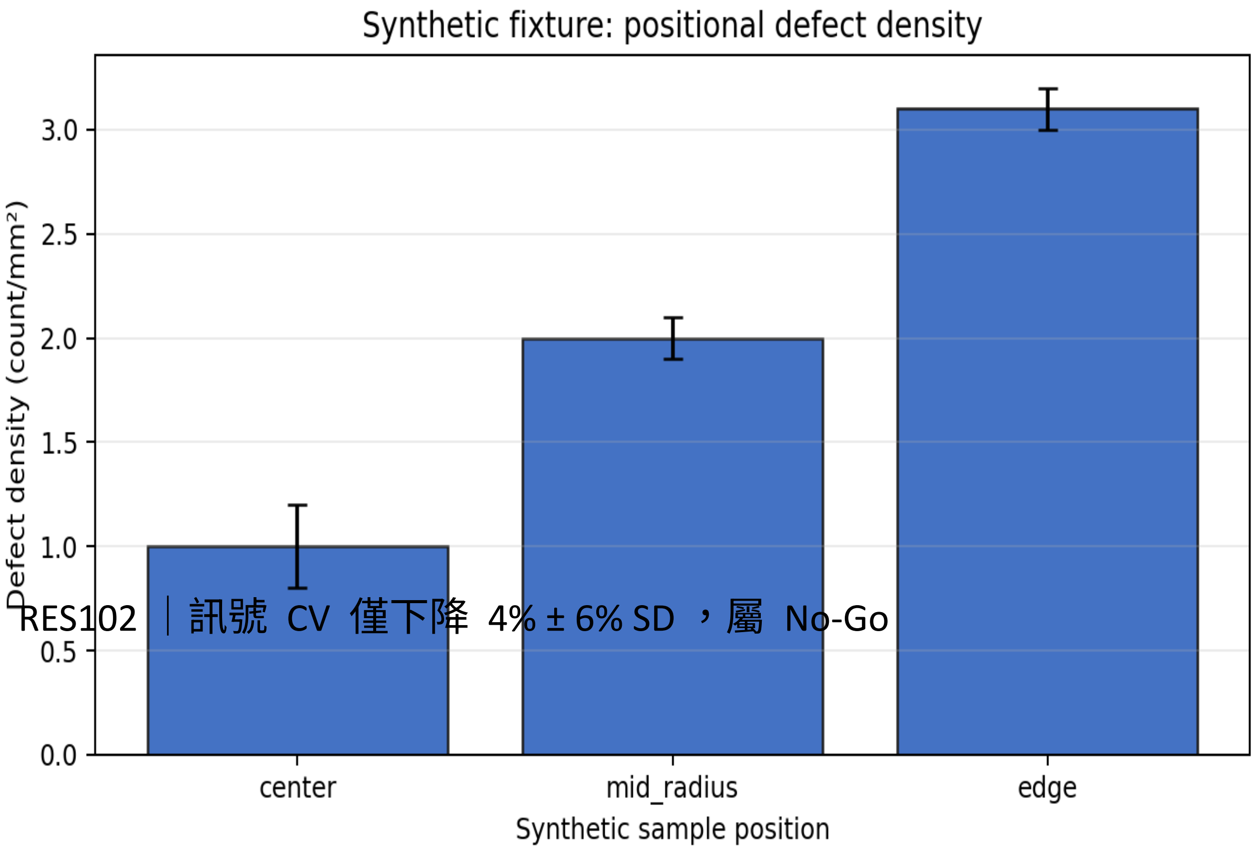
RES101 ｜平均導電度增加 24% ± 5% SD



RES101 ｜平均導電度增加 24% ± 5% SD

H001 ｜ Result

RES102 ｜訊號 CV 僅下降 4% ± 6% SD ，屬 No-Go



RES102 ｜訊號 CV 僅下降 4% ± 6% SD ，屬 No-Go

H001 ｜ Integrated Discussion

RES101

RES102

Mechanism assessment ｜ bulk conductivity 是部分機制，不足以解釋 interface-sensitive variation 。
Alternatives ｜接觸電阻；局部壓力差；電極幾何
Remaining uncertainty ｜ contact interface 是否為主導因素仍需匹配導電度後檢驗。

H001 ｜ Layer Summary / Decision

Answered ｜ bulk conductivity 影響缺陷梯度，但不足以單獨解釋重複性。
Hypothesis status ｜ partial_support
Decision ｜ Partial-Go: Bulk conductivity is contributory but insufficient.

interface contact 的貢獻尚未量化。

Next question ｜匹配 bulk conductivity 後， contact resistance 是否主導變異？
Next Step ｜ NS101

H001 ｜ Hypothesis Transition

Previous hypothesis ｜ C101
Key results ｜ RES101, RES102
New observation ｜ E104
New hypothesis ｜ C201

Not explained ｜導電度增加後訊號重複性沒有改善。
Therefore ｜剩餘變異隨電極接觸位置改變，因此核心解釋由 bulk transport 轉向 interface contact 。

H002 ｜ Hypothesis

Hypothesis ｜ Contact resistance drives instability at low pressure.
Falsifier ｜ Matched conditions do not change the predicted CV or resistance.
Research question ｜低接觸壓力下， contact resistance 是否主導訊號不穩定？

H002 │ Problem

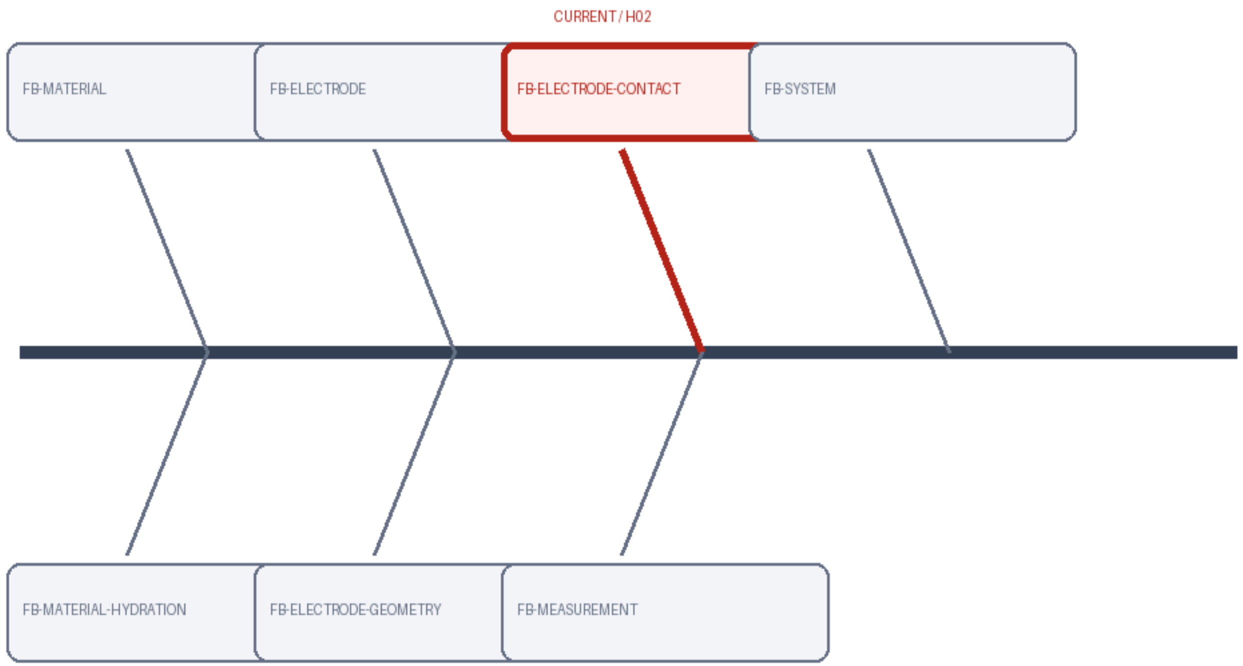
H01 支持導電度梯度存在 ; H01 未支持導電度單獨主導重複性

bulk transport 與 interface contact 的相對貢獻尚未區分。

低接觸壓力下， contact resistance 是否主導訊號不穩定？

H002 | Total Fishbone / Research Map

FB-ELECTRODE-CONTACT



H002 ｜ Observation → Literature → Mechanism

Observation ｜位置依賴缺陷與訊號變異已觀察到。
Problem ｜現有模型無法解釋此變異。

Literature consensus ｜ transport gradient 可產生位置效應。
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Mechanism ｜ contact resistance

匹配導電度並改變接觸壓力

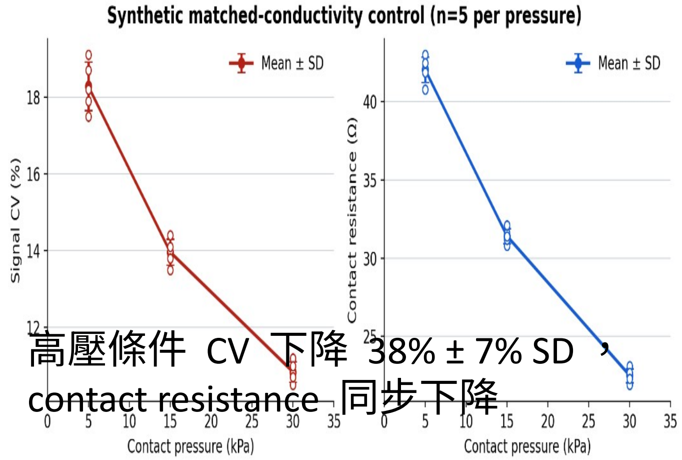
低接觸壓力下， contact resistance 是否主導訊號不穩定？

Registered observation visual.

H002 | Results

IV: contact pressure
Controls: bulk conductivity; 電極幾何
Baselines: low-pressure control
N/replicates: replicates: 5; samples: 15
Metrics/units: 訊號 CV (%); contact resistance (ohm)
Method: synthetic-pressure-fixture
Prediction: CV and resistance decrease

go: both metrics improve;
partial_go: one metric improves;
no_go: neither improves



高壓條件 CV 下降 38% ± 7% SD ， contact resistance 同步下降

高壓條件 CV 下降 38% ± 7% SD ， contact resistance 同步下降

H002 ｜ Layer Summary / Decision

RES201

None identified

Cross-experiment pattern ｜匹配導電度後，提高接觸壓力明顯降低變異。
Mechanism assessment ｜ contact resistance 在低壓條件下主導不穩定。
Alternatives ｜電極幾何仍可能調節壓力分布

Discussion uncertainty ｜高壓長期循環是否造成新的 wear mechanism 尚未知。
Summary unresolved ｜高壓循環的耐久性與 wear 尚未測試。

Hypothesis status ｜ supported
Decision ｜ Go: Contact-pressure control discriminates the interface mechanism.

Next question ｜接觸壓力改善能否在長期循環下維持？
Next Step ｜ NS201